

ASSIGNMENT : 4

1. A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

Ans: -

Laptop Won't Power On – Troubleshooting Steps

1. Check the Power Adapter

- Make sure the charger is properly connected to the laptop and wall socket.
- **Checking:** Damaged cable, loose connection, or faulty adapter.

2. Check the Power Outlet

- Plug another device into the same outlet.
- **Checking:** Whether the wall socket is supplying power.

3. Check the Charging Indicator

- Connect the charger and look for the charging/power LED.
- **Checking:** Whether the laptop is receiving power.

4. Perform a Power Reset

- Disconnect the charger, remove the battery if removable, hold the power button for **15–30 seconds**, then reconnect the charger and try again.
- **Checking:** Temporary power or hardware-state problems.

5. Check Internal Hardware

- If there is still no response, inspect the battery, RAM, charging port, and motherboard connections.
- **Checking:** Loose, damaged, or faulty hardware causing the no-power condition.

2. You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Hint: Consider checking both hardware connections and basic display settings.

Ans: -

1. **Check Monitor Power**

- Make sure the monitor is switched ON and the power LED is lit.

2. **Check Display Cable**

- Check HDMI/Display Port/VGA cable connections at both the monitor and PC.

3. **Select Correct Input Source**

- Use the monitor buttons to select the correct **HDMI/DP/VGA input**.

4. **Check PC Display Port**

- If a dedicated GPU is installed, connect the monitor cable to the **GPU port**, not the motherboard port.

5. **Restart the PC**

- Restart the computer and check whether the display returns.

6. **Test Another Cable/Monitor**

- Use a known-working cable or monitor to identify whether the problem is with the display equipment.

7. **Check RAM/GPU**

- Power off the PC and reseal the RAM and GPU if there is still no display.

8. **Check Display Settings**

- If Windows starts but the screen remains blank, press **Win + P** and select **PC screen only** or **Duplicate**.

9. **Confirm the Fix**

- Verify that the monitor displays the desktop normally and remains stable after restarting.

3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

Ans: -

1. **Check Mouse Power**

- Replace the batteries or charge the mouse.
- **Check:** Whether the mouse has enough power.

2. **Check Wireless Receiver**

- Remove and reconnect the USB receiver.
- **Check:** Whether the receiver is properly connected.

3. **Test Another USB Port**

- Connect the receiver to a different USB port.
- **Check:** Whether the original USB port is faulty.

4. **Test the Mouse on Another PC**

- Connect the mouse/receiver to another computer.
- **Check:** If the mouse works, the original system may have a settings/driver issue.

5. **Check Windows Settings**

- Go to **Settings** → **Bluetooth & devices** → **Mouse** and check that the mouse is detected.
- **Check:** Mouse configuration and connection settings.

6. **Check Device Manager**

- Open **Device Manager** → **Mice and other pointing devices**.
- **Check:** Driver errors or disabled mouse devices.

7. **Confirm the Cause**

- If another mouse works on the same USB port, the original mouse is likely faulty.
- If no mouse works on the port, check the **USB port/system**.
- If the mouse works elsewhere but not on this PC, investigate **system settings or drivers**.

4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Hint: Think about fans, dust, and thermal paste.

Ans: -

1. **CPU/GPU Fan Not Working**

- **Check:** Open the case and observe whether the CPU and GPU fans are spinning.
- **Solution:** Clean, reconnect, or replace the faulty fan.

2. **Dust Build-up**

- **Check:** Inspect the CPU heatsink, GPU, case fans, and air vents for dust.
- **Solution:** Clean the dust using compressed air.

3. **Old or Dried Thermal Paste**

- **Check:** Inspect the thermal paste between the CPU and heatsink.
- **Solution:** Clean the old paste and apply fresh thermal paste properly.